TriTF: A Triplet Transformer Framework Based on Parents and Brother Attention for Hyperspectral Image Change Detection

Xianghai Wang, Keyun Zhao, Xiaoyang Zhao, and Siyao Li

***Abstract*— Hyperspectral image change detection (HSI-CD) is a technique to accurately detect land cover changes by using HSIs with rich spatial–spectral information. In recent years, the HSI-CD methods based on convolutional neural networks (CNNs) have achieved great success because of their flexible and effective feature extraction ability. However, these methods often take the HSI patches as the input of the networks, which undoubtedly hinders the overall perception of the HSIs. Meanwhile, the valuable temporal information in HSIs is often underutilized. For this end, a triplet transformer framework (TriTF) based on parents-temporal attention (PTA) and brother-spatial attention (BSA) is proposed for HSI-CD. The proposed framework mainly contains the following three parts: 1) transformer-based network backbone, which uses the self-attention to capture the correlation between arbitrarily two pixels in the same patch and extracts the global spatial correlation in the unit of encoded input patches; 2) PTA branch; unlike the previous cross-temporal attention mechanisms of the "T1↔T2" mode which only consider the interaction between bitemporal HSIs, this article constructs a novel PTA of the "T1→T3←T2" mode, which takes the difference-temporal image T3 as the core and the impact of bitemporal HSIs on the land cover changes is more concerned in the PTA; and 3) BSA branch. The most similar patch in the current training batch of each patch is defined as its brother patch. Furthermore, cross-spatial attention is applied to propagate the features of the brother patch to the current patch. Thus, the middle- and long-range dependencies can be utilized and the scope of feature propagation can be extended. In this article, the experiments under low and high sampling rates are conducted and proved the outstanding change detection performance of the proposed TriTF when compared with abundant state-of-the-art (SOTA) CD algorithms. The source code of this article will be released at https://github.com/zkylnnu/TriTF.**

***Index Terms*— Change detection (CD), deep learning, hyperspectral image (HSI), self-attention, transformer.**

I. Introduction

HYPERSPECTRAL images (HSIs) are a kind of remote sensing images composed of hundreds of spectral channels captured by imaging spectrometers. Because of the unique spectral curves of ground objects with different components, remote sensing images with high spectral resolution have obvious advantages for fine observation of land cover and land use. At present, HSIs have been used in many fields of Earth observation. Among them, hyperspectral image change detection (HSI-CD), as an important technology to identify and analyze the changes between time series HSIs obtained in the same geographic area [1], has been widely used in agriculture and forestry monitoring, water resources monitoring, urban change analysis, and many other domains [2], [3], [4]. Change detection (CD) techniques have developed greatly over the past few decades. Generally, the traditional HSI-CD methods can be roughly divided into four categories: algebra-based methods, transformation-based methods, classification-based methods, and other advanced methods.

Early algebra-based methods, such as image difference, image ratio, and image regression, mainly determine the changed area by straightforward algebraic manipulation [5]. In particular, change vector analysis (CVA) is the most commonly used algebra-based method, which identifies the binary change information through spectral vector subtraction and threshold calculation [6]. However, the detection accuracy of these methods relies heavily on the quality of radiometric and geometric correction; transformation-based methods often transform the image into a specific feature space to highlight the changed information and suppress the unchanged information [7], such as principal component analysis (PCA) [8], subspace-based CD [9], and iteratively reweighted multivariate alteration detection [10]. Although such methods have a good performance in reducing spectral redundancy, they suffer a certain loss of spectral information and perform poorly on class-imbalanced tasks; According to the different operation processes, the classification-based methods can be mainly divided into direct and postclassification methods [11], [12]. The direct classification methods usually combine multitemporal HSIs into one and then use a classifier to detect the changed categories. The postclassification methods first classify the multitemporal images individually and analyze the changes according to the classification results. Actually, these methods can overcome the problem of sensor inconsistency, but the CD performance depends heavily on the classification accuracy [13].

In addition to the above methods, there are also outstanding traditional methods to approach HSI-CD tasks from different

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Xianghai Wang is with the School of Geography and the School of Computer and Information Technology, Liaoning Normal University, Dalian 116029, China (e-mail: xhwang@lnnu.edu.cn).

Keyun Zhao and Siyao Li are with the School of Computer and Information Technology, Liaoning Normal University, Dalian 116029, China.

Xiaoyang Zhao is with the School of Geography, Liaoning Normal University, Dalian 116029, China.

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perspectives. For example, the unmixing-based methods can utilize the subpixel level information in HSIs through endmember extraction and abundance estimation [14], [15]. The tensor-based methods can preserve the structural characteristics of HSIs more effectively and make use of them [16], [17]. The spectral analysis-based methods pay more attention to the spectral variations and give full play to the advantage of the high spectral resolution of HSIs [18], [19]. The morphological-based methods can effectively utilize the intrinsic local pixel relationship [20].

Currently, the HSI-CD methods based on deep learning gradually occupy a dominant position in this field due to their powerful representational learning ability and significantly improve the degree of automation compared with the traditional methods [21]. At the present stage, the widely used deep learning methods mainly include recurrent neural networks (RNNs) [22], convolutional neural networks (CNNs) [23], and graph neural networks (GNNs) [24]. All of them have been successfully applied in the field of HSI-CD and obtained certain advantages. For example, Yuan and Li [25] proposed a dual partial recurrent network that combined recurrent network, unsupervised slow feature analysis, and CVA; Ou et al. [26] started with band selection and proposed a framework based on slow–fast band selection and feature fusion grouping (SFBS-FFGNET) for HSI-CD; and Qu et al. [27] introduced graph convolutional network into HSI-CD and proposed a dual-branch difference amplification graph convolutional network, which can maintain favorable CD performance with low sampling rates.

In recent years, the attention mechanism, with its powerful modeling ability and strong versatility, has become a widely used and effective feature extraction method in the field of HSI-CD and is usually used to enhance the feature expression ability of CNNs. For instance, Yang et al. [28] proposed a deep multiscale pyramid network enhanced with spatial–spectral residual attention (DMPs^{2}raN). The multiscale pyramid convolution can effectively excavate multilevel and multiscale spatial–spectral features and the features were further enhanced by a spatial–spectral residual attention module of each scale; Wang et al. [29] proposed a spectral–spatial-wise attention-based Siamese network (SSA-SiamNet), which can emphasize the channels and spatial locations with rich information and suppress the channels and spatial locations with less valuable information, so as to adaptively refine the spectral–spatial features. In summary, although these attention-based methods have achieved superior detection performance, there are still many problems that have not been well solved by the existing HSI-CD methods.

On the one hand, limited by the data size of HSIs and the patchwise input of the networks, most of the existing deep learning-based CD networks are devoted to reasonably introducing the attention mechanism to obtain a dense affinity matrix to encode the local spatial correlation within a patch. However, the utilization of middle- and long-range dependencies in HSIs is obviously insufficient, while the effectiveness of them has been demonstrated in many fields [27], [30]. On the other hand, in most HSI-CD methods, the bitemporal HSIs are often first processed by concatenation or subtraction operation and input to the network as a whole [26], [27]. Therefore, the temporal dependencies in the HSIs are almost completely ignored. Some other methods capture the dependencies between bitemporal HSIs by cross-temporal interaction attention, which realizes the cross-temporal feature propagation by constructing the affinity relationship between the corresponding patches of bitemporal HSIs. However, this construction method can only propagate the change information between HSIs but cannot capture and emphasize it directly. Thus, it is still difficult to make full use of temporal dependencies to enhance the discriminant ability of the network to change information.

Based on the above issues, a triplet transformer framework (TriTF) based on parents-temporal attention (PTA) and brother-spatial attention (BSA) named TriTF is proposed in this article. The main contributions of this work are given as follows.

1) A novel end-to-end transformer framework named TriTF is proposed for HSI-CD, which adopts a "branch-backbone-branch" structure. In this framework, the spatial–spectral information and temporal information in the HSIs can be fully mined. Abundant experiments on three public datasets proved the superiority of TriTF when compared with state-of-the-art (SOTA) methods.
2) The PTA is designed to mine the temporal change information, which is a ternary attention module of the "T1$\rightarrow$T3$\leftarrow$T2" mode. Unlike cross-temporal interactive attention, PTA can better model the correlations between land cover and surface changes, thus enhancing the feature expression of the network. Temporal information to enhance the feature expression performance of the network.
3) The BSA is designed to utilize the middle- and long-range dependencies of the land cover, which breaks the receptive field limitation of the patchwise unit. BSA adopts cross-spatial attention to exploit the correlations between land cover. In this way, BSA can capture more learnable information and achieve more valuable feature propagation.

The rest of this article contains the following sections. Section II introduces the related work, involving the applications of transformer structure in HSI processing, the study of temporal dependencies between HSIs, and the exploration of middle- and long-range dependencies in land cover. Section III describes the proposed HSI-CD method TriTF in detail. Section IV introduces the comparison experiments and ablation study. Section V contains comprehensive comparison experiments with multiple sampling rates, the analysis of the influence of some important parameters on the performance of the proposed method, and the complexity efficiency analysis. Section VI draws the conclusion.

II. Related Work

A. Transformer in HSI Processing

The transformer is a model structure that relies entirely on multihead attention mechanisms to construct global dependencies between inputs and outputs [30]. This structure has

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attracted huge attention since it was proposed. Although the transformer prototype was used for natural language processing tasks, its excellent feature extraction and generalization capabilities make it successfully applied in many other fields, including HSI processing.

Hong et al. [31] purely used the transformer structure for the HSI classification task without any preprocessing operation for the first time and proposed a new transformer-based backbone named SpectralFormer. In the SpectralFormer, cross-layer adaptive fusion and groupwise spectral embedding were used for learning locally detailed spectral representations and the transmission of memory-like components from shallow to deep layers. Sufficient experiments proved the superior performance of SpectralFormer compared with other methods. Yu et al. [32] proposed a multilevel spectral–spatial transformer network (MSTNet) for the HSI classification task, which used imagewise input and transformer to extract global information and fused multilevel features to enhance the expression of HSI. Sun et al. [33] proposed a spectral–spatial feature tokenization transformer (SSFTT) for HSI classification. The SSFTT first extracted spatial–spectral features by using plain CNNs. Then, the feature transformation process was completed by a Gaussian weighted feature tokenizer. Finally, the extracted features were input to the transformer encoder module. Hu et al. [34] proposed a transformer-based framework named Fusformer for HSI super-resolution, which was the first application of transformers in the field of HSI super-resolution. Overall, transformer structure was more widely applied in the field of HSI classification and less in the application of HSI-CD. Wang et al. [35] first introduced the transformer to HSI-CD and proposed a spectral–spatial–temporal transformer (SST-Former). In SST-Former, each pixel in the HSI cube was embedded to record the spectral location and spatial location. Then, the spatial transformer and spectral transformer were applied to extract the sequence information of spectra and the spatial texture information. Finally, the temporal transformer was used to extract useful CD features between the HSI pairs.

In summary, transformer structure is a very effective feature extraction method, and its application and exploration in the HSI-CD field are still in the initial stage. Therefore, how to organically combine the characteristics of HSI-CD tasks with transformers is still a very valuable research topic.

B. Temporal Dependencies in Multitemporal HSIs

The mining and utilization of temporal dependencies are key points of time series analysis tasks. For HSI-CD tasks, it aims to use the cross-temporal correlation between multitemporal HSIs to guide the capture of land surface change information, which is also one of the biggest differences compared with the HSI classification task. In essence, the necessity and effectiveness of temporal dependencies are closely related to the physical significance of the CD task. Since the multitemporal HSIs sequence is a direct representation of land cover change in a certain area and can be considered as a process of geographical evolution in the temporal dimension, the correlation of land cover in the temporal dimension is a potential indicator of land cover change [36].

At present, a few scholars have explored the temporal dependencies in HSIs, and Song et al. [37] presented a cross-temporal symmetric attention network (CSANet), which applied cross-temporal attention mechanism to capture the temporal correlations between the time series HSIs and combined them with spatial attention and spectral attention. In a similar way, Wang et al. [35] also used cross-temporal attention to model the temporal dependencies, which were extracted from the spatial–spectral features instead of the original HSIs. Thus, the attention mechanism focused more on the relationship between two pairs of features. Shi et al. [38] used a multipath convolutional long short-term memory (MP-ConvLSTM) with the parallel structure to learn the multilevel temporal dependencies in the HSIs and a Siamese CNN to extract the preliminary features. Although temporal dependence is a characteristic of the HSI-CD task and has been proven to be effective, it is still poorly explored and ignored by most methods. Therefore, more in-depth explorations of temporal dependencies have important practical significance.

C. Middle- and Long-Range Dependencies in HSIs

The middle- and long-range dependencies play an important role in the field of image processing, which indicates that the network can perceive information in a large scope. Different from the local correlation with less information perception, the middle- and long-range dependencies can represent the relationship between any two pixels with a long distance in the image and provide more abundant learnable knowledge for the network. In the field of HSI processing, the middle- and long-range dependencies can make up for the deficiency of local dependencies in modeling land cover relationships. For example, Hong et al. [39] modeled the dependencies between all land cover in HSI by adjacency matrix and adopted graph convolution to propagate features in topological space. This approach achieved good classification results, but the computational complexity was very high. Yu et al. [32] adopted an imagewise input instead of patchwise input and modeled global dependencies through global position encoding. This was one of the most straightforward ways to use a transformer to leverage middle- and long-range dependencies, but too many learnable parameters can lead to high complexity and poor performance when a few labeled samples were available. In the field of HSI-CD, Qu et al. [27] used graph convolution to transfer the feature propagation process to the topological space and used an adjacency matrix to model the long-range dependencies between superpixel-level graph nodes, which had lower complexity compared with the pixel-level graph structure but also lost the ability to detect fine changes.

In general, there is still little research on middle- and long-range dependencies in the field of HSI-CD. We believe that the main reasons are given as follows. First, most CNN-based methods cut HSIs into patches with a small spatial size and take them as the input of the network. Although such processing modes can effectively alleviate the problem of insufficient samples and high network computational complexity, they also undoubtedly limited the receptive field of the network. Besides, a few methods adopted imagewise

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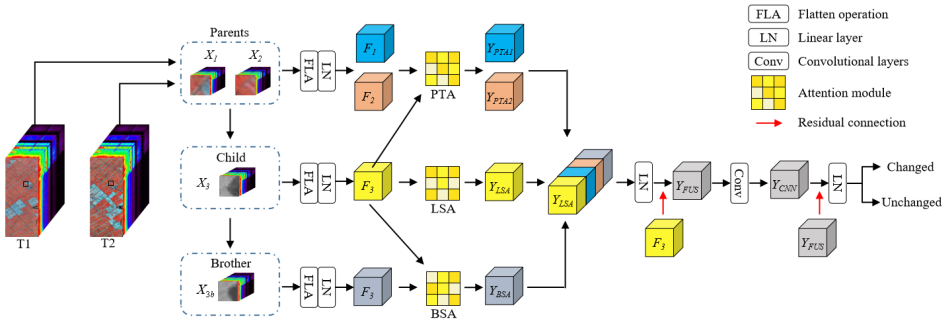


Fig. 1. Brief architecture of the proposed TriTF.

input and conducted global correlation modeling at pixel level or superpixel level. However, it was difficult to achieve a balance between computational complexity and detection performance. Therefore, how to effectively utilize the middle- and long-range dependencies between land cover in HSIs remains an urgent problem to be solved.

III. Proposed Method

This section describes the overall structure of the proposed TriTF and the detailed design of each module, including the network backbone, the main innovation modules in the two branches, and the prediction and loss function. The brief architecture of TriTF is shown in Fig. 1.

First, 3-D HSI patches are generated from the original bitemporal HSIs T1,T2 $\in \mathbb{R}^{H\times W\times C}$, where H, W, and C denote the height, width, and the number of channels of HSIs, respectively. Also, the extracted HSI patches are defined as X_1, $X_2 \in \mathbb{R}^{p\times p\times C}$, where p denotes the width and height of the patches (namely, the hyperparameter patch size). Second, subtract X_1 and X_2 to get the difference-temporal HSI patch X_3 (since X_3 is generated from X_1 and X_2, we call them parents of X_3); Then, we search for the most similar patch in this epoch of training for the difference-temporal HSI patch X_3, namely, brother patch X_{3b}. This similarity is measured according to the Euclidean distance between patches. The search scope is limited to the current batch, that is, the patch with the smallest Euclidean distance is searched among the number of B patches (B denotes the batch size). So far, the input of the network has been obtained. Input X_1 and X_2 to the PTA branch, X_3 to the network backbone, and X_{3b} to the BSA branch. In the network backbone (see Section III-A), local spatial attention (LSA) models the global dependencies between pixels within patch. The local spatial information of the difference-temporal HSI can be adequately utilized by the multihead self-attention mechanism in LSA. In the PTA branch, PTA models the dependencies between bitemporal land cover and land cover change information by a triplet temporal attention and then propagates land cover information to the difference-temporal HSI according to the temporal dependencies. In the BSA branch, BSA can construct the correlation between the difference-temporal patch and its brother patch, that is, the middle- and long-range dependencies. The valuable information in the brother patch is propagated to the current patch by using cross-spatial attention. Finally, the features extracted by the two branches are fused to the network backbone and the plain CNNs are adopted to extract features from the fusion features, and the network output is obtained through the fully connected layers.

A. Transformer-Based Backbone

The transformer structure adopts the self-attention mechanism and pointwise, fully connected layers to constitute both the encoder and the decoder. The encoder is made up of six identical layers, and each layer contains a multihead attention mechanism and a pointwise fully connected feedforward network. Differently, in addition to the above two elements, the decoder also contains masked multihead attention, which is used to ensure that the predictions are unidirectional in natural language processing tasks [30].

Actually, the effectiveness of the transformer depends mainly on the use of multihead attention, which is formed by multiple self-attention layers. The self-attention mechanism pays more attention to the internal correlations of features rather than the correlations between features, thus effectively reducing dependence on external information. Therefore, to better capture the internal correlations of HSI patches, we take the encoder of the transformer structure as the network backbone, as shown in Fig. 2.

1) Data Preprocessing: First, the bitemporal HSIs are cut into patches and each patch corresponds to a real label, that is, changed or unchanged. The input of backbone is the difference-temporal patch $X_3 \in \mathbb{R}^{p\times p\times C}$. In order to facilitate the processing of transformer structure, similar to [40] and [41], this article first performs flatten operation on the patches to compress its spatial dimension. Then, in order to avoid the Hughes phenomenon of HSIs, we use the fully connected layers to reduce the dimension of the input to c ($c < C$). The features after dimensional reduction are defined as $F_3 \in \mathbb{R}^{M\times c}$, where $M = p \times p$. Also, F_3 is used as the input of LSA.

2) Local Spatial Attention: The structure of LSA is shown in Fig. 3. In order to stabilize the performance of the network, this article adopts the multihead attention mechanism. First, $3n$ (n is the number of the attention heads) linear layers of the same size are applied to preprocess the input features, and Q^3, K^3, $V^3 \in \mathbb{R}^{n\times M\times g}$ are generated, where $g = c/n$ denotes the number of channels of each attention head, and Q^3, K^3, and V^3 denote the queries, keys, and values of X_3, respectively.

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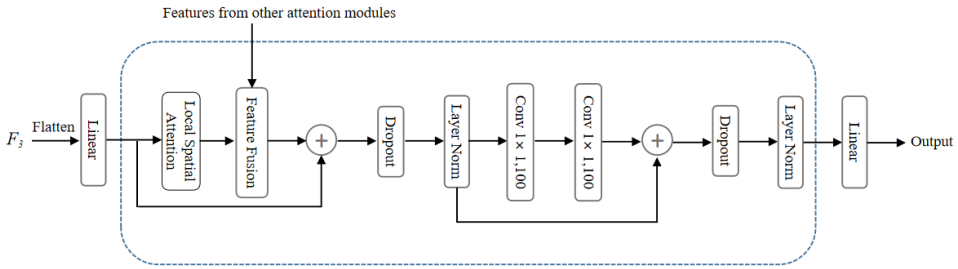


Fig. 2. Transformer-based backbone in TriTF.

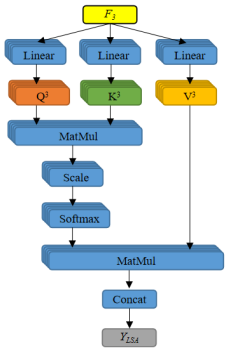


Fig. 3. LSA.

Then, the LSA is applied to propagate the features, which can be defined as follows:

$$\begin{aligned} Y_{\mathrm{LSA}} &= \mathrm{LSA}\left(Q^3, K^3, V^3\right) \\ &= \mathrm{Concat}|_{i=1}^{n}\, \mathrm{softmax}\left(\frac{Q_i^3\left(K_i^3\right)^T}{\sqrt{d_{k^3}}}\right) V_i^3 \end{aligned} \tag{1}$$

where $(K_i^3)^T$ denotes the transposed matrix of K_i^3, d_{k^3} denotes the number of channels of the input K^3, Concat denotes the concatenate operation, and softmax is the activation function. The final features obtained by LSA are defined as $Y_{\mathrm{LSA}} \in \mathbb{R}^{M\times c}$.

3) Feature Fusion: After LSA is executed, the output Y_{LSA} is fused with the output features of other branches, they are Y_{PTA1}, Y_{PTA2}, and Y_{BSA} (see Sections III-B and III-C). Next, linear layers are applied to reduce the channels to c. Then, the dimensional reduction result is concatenated with the feature F_3, which is the feature before LSA, and this represents a residual connection. Finally, the fusion feature $Y_{\mathrm{Fus}} \in \mathbb{R}^{M\times c}$ is obtained and these processes can be expressed as follows:

$$Y_{\mathrm{Fus}} = F_3 + \mathrm{Linear}(\mathrm{Concat}(Y_{\mathrm{LSA}}, Y_{\mathrm{PTA1}}, Y_{\mathrm{PTA2}}, Y_{\mathrm{BSA}})) \tag{2}$$

where Linear denotes the linear layers, in which the input and output channels are $4c$ and c, respectively.

4) CNN Layers: After the feature fusion, two CNN layers are applied to improve the ability of the network to express local features. The kernel size of CNN layers is 1×1 and the numbers of kernels are the same. Besides, the ReLU activation function is used between the two CNN layers. These processes are defined as follows:

$$Y_{\mathrm{CNN}} = u_i^2 * \mathrm{ReLU}\left(u_i^1 * Y_{\mathrm{Fus}} + b_i^1\right) + b_i^2 \tag{3}$$

where $*$ denotes the convolution operation, u_i^1 and u_i^2 denote the ith kernel of the first layer and second layer, respectively, and b_i^1 and b_i^2 denote the corresponding biases, in which i ranges from 1 to c. Finally, in order to enable the network to extract deep-level features, the aforementioned transformer-based backbone loops r times (r is a hyperparameter that denotes the network depth and is set to 2), that is, taking the output of the first backbone as the input of the second backbone. The output of the second backbone is defined as Y_{OUT}.

B. Parents-Temporal Attention

Considering the necessity of temporal dependencies in HSI-CD, this article designs a PTA branch for the utilization of temporal dependencies. Unlike the "T1↔T2" mode cross-temporal attention which is used by most methods [35], [37], PTA is ternary temporal attention, which introduces the difference-temporal features. The initial inputs of this branch are X_1, $X_2 \in \mathbb{R}^{p\times p\times C}$. As with the network backbone, the inputs are first flattened and dimensionally reduced. Then, the features after dimension reduction F_1, $F_2 \in \mathbb{R}^{M\times c}$ and the backbone features F_3 are input to PTA together for cross-temporal feature propagation, as shown in Fig. 4. The PTA process can be defined as follows:

$$\begin{cases} Y_{\mathrm{PTA1}} = \mathrm{PTA}\left(Q^1, K^3, V^1\right) \\ \qquad = \mathrm{Concat}|_{i=1}^{n}\, \mathrm{softmax}\left(\dfrac{Q_i^1\left(K_i^3\right)^T}{\sqrt{d_{k^3}}}\right) V_i^1 \\ Y_{\mathrm{PTA2}} = \mathrm{PTA}\left(Q^2, K^3, V^2\right) \\ \qquad = \mathrm{Concat}|_{i=1}^{n}\, \mathrm{softmax}\left(\dfrac{Q_i^2\left(K_i^3\right)^T}{\sqrt{d_{k^3}}}\right) V_i^2 \end{cases} \tag{4}$$

where Q^1 and Q^2 denote the queries of temporal 1 and 2, and they are obtained by weight-sharing linear layers. V^1 and V^2 denote the corresponding values. K^3 is obtained in Section III-A2. The responsiveness of temporal 1 and 2 to change information can be obtained by PTA and it can be utilized by the subsequent feature fusion.

C. Brother-Spatial Attention

The middle- and long-range dependencies between land cover are of great value in Earth observation tasks, which provide more reference information for training tasks under

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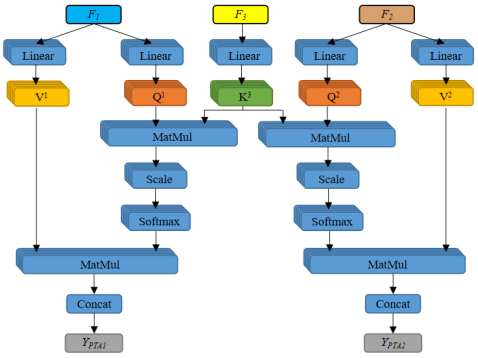


Fig. 4. PTA.

the condition of insufficient information. However, they are often ignored in the HSI processing field. Since most of the existing deep learning-based HSI-CD methods divide the original HSIs into patches of small size before network input and the subsequent feature extraction process is spread out within the patch, it seems difficult to explore the middle- and long-range dependencies in this paradigm. Considering this issue, this article designs the BSA that can effectively utilize the middle- and long-range dependencies between land cover in the condition of patchwise input, as shown in Fig. 5.

First, we calculate the average pixel values of patches by spatial location in each training batch. Second, the similarities between patches are calculated according to the following equation:

$$S_{m,n} = -\sum_{j=1}^{M}\left(x_j^m - x_j^n\right)^2 \tag{5}$$

where $S_{m,n}$ denotes the similarity between the mth patch and the nth patch ($m \neq n$) and x_j^m and x_j^n denote the average pixel values of the mth patch and the nth patch at the jth spatial location, respectively ($j = 1, 2, \ldots, M$). Then, the brother patch of each patch can be determined according to the similarity matrix S. The attention process of BSA can be represented as follows:

$$\begin{aligned} Y_{\text{BSA}} &= \text{BSA}\left(Q^{3b}, K^3, V^{3b}\right) \\ &= \text{Concat}|_{i=1}^{n}\ \text{softmax}\left(\frac{Q_i^{3b}\left(K_i^3\right)^T}{\sqrt{d_{k^3}}}\right)V_i^{3b} \end{aligned} \tag{6}$$

where Q^{3b} and V^{3b} denote the query and value of the brother patch, respectively. In particular, the dependencies between patches could be arbitrary range and the brother patch of each patch varies in different epochs. Thus, BSA enables a certain patch to aggregate the features of many different patches so that the utilization factor of information is improved and the middle- and long-range dependencies between land cover are adequately played.

To visually show the different working scopes of different attention mechanisms, we take Fig. 6 as an example for a simple illustration. The first one is LSA, which works on difference-temporal image T3. The action scope of LSA is the interior of patches, as p_0 shown in Fig. 6. LSA focuses on the local information in the patches and achieves the feature propagation within the patch range. The second one is PTA and it processes all three temporal HSIs. PTA works on the patches belonging to different temporal HSIs that are located in the same geographical location, that is, p_0, p_1, and p_2 in Fig. 6. Besides, PTA focuses on the influence of bitemporal HSI T1 and T2 to the difference-temporal image T3 and realizes the cross-temporal feature propagation. The last one is BSA, which processes the difference-temporal image T3. BSA works on the patches and their brother patches, which belong to the same HSI and different geographical locations, just like p_0 and p_3 in Fig. 6. Certainly, BSA can effectively utilize the middle- and long-range dependencies between land covers to achieve valuable cross-spatial feature propagation between patches.

D. Prediction and Loss Function

Through the above processes, the network finally obtains the feature map $Y_{\text{OUT}} \in \mathbb{R}^{M\times c}$ (see Section III-A4). Then, we compress the channels and achieve the predictions through a fully connected layer. Therefore, the predictions of the network for the current batch of samples can be expressed as $Y_{PR} \in \mathbb{R}^{B\times 2}$, where B denotes the batch size.

In deep learning tasks, our goal is to obtain the predictions of the network infinitely close to the ground truth, that is, to minimize the difference between them. Thus, the loss functions are adopted to implement this process, which is used to calculate the difference value between predictions and ground truth, thus providing the input value for backpropagation. To this end, it is quite crucial to select a suitable loss function for network optimization during the training phase. Considering that the CD task can be regarded as a binary classification task (changed or unchanged), we adopt the binary cross-entropy loss function to optimize our network, which is commonly used in classification problems. Meanwhile, we update the network parameters once per batch rather than once per epoch. Thus, the loss function calculation formula of our method can be expressed as

$$L_{\text{BCE}} = -\sum_{i=1}^{B}\left(Y_{GT}^i \log Y_{PR}^i + \left(1 - Y_{GT}^i\right)\log\left(1 - Y_{PR}^i\right)\right) \tag{7}$$

where Y_{GT} denotes the one-hot ground-truth map corresponding to the input patches in the current training batch and Y_{GT}^i is the ith element of it; Y_{PR}^i represents the ith element of the network predictions Y_{PR}. Certainly, the more precise the CD results of the network, the lower the value of L_{BCE}.

IV. Experiments and Analysis

A. Datasets and Experimental Setup

To verify the effectiveness of the proposed method, three widely used public datasets are used for the experiments, namely, River [42], Farmland [43], and USA [44] datasets. The pseudo-color images of three datasets are shown in Fig. 7, and their detailed information is shown in Table I.

The River dataset is captured in Jiangsu, China, by the Earth Observing-1 (EO-1) Hyperion sensor. The spatial resolution of the River dataset is 30 m, and the changes in this dataset

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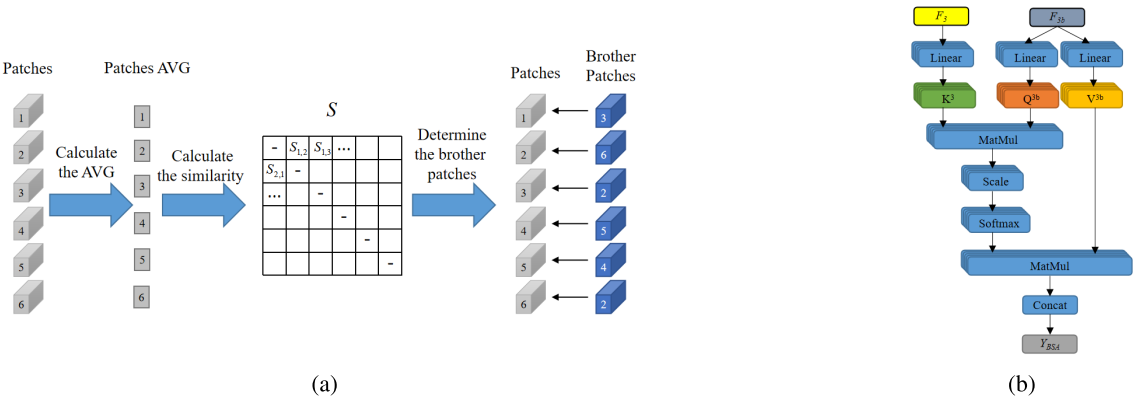


Fig. 5. BSA. (a) Search process of brother patches (the six patches represent the patches in one training batch). (b) Attention process.

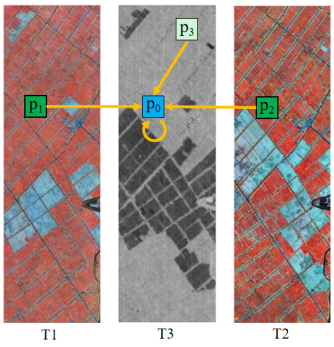


Fig. 6. Schematic of the scopes of different attention mechanisms.

TABLE I
DATASET INFO

Dataset	River	Farmland	USA
Size	463×241	450×140	307×241
Channels	198	155	154
Changed	9698	18277	16676
Unchanged	101885	44723	57311
Date of $T1$	May 3, 2013	May 3, 2006	May 1, 2004
Date of $T2$	Dec 31, 2013	Apr 23, 2007	May 8, 2007

are mainly related to the sediments. In particular, there is a serious class imbalance in this dataset and the changes are subtle; thus, this dataset is quite difficult to detect. The second dataset, named Farmland, contains a farmland area in Yancheng, Jiangsu, China. The bitemporal HSIs are also acquired by the EO-1 Hyperion sensor. The changes in this dataset mainly reflect the cultivation area and crop changes, which are relatively concentrated. The third dataset, named the USA, mainly contains soil, irrigation areas, rivers, and other terrains. The sensor that captured this dataset is also EO-1 Hyperion, and the changes in this dataset are mainly the large cultivated land changes and relatively fine riverbank line changes.

To facilitate the comparison of different methods, this article adopts the two most commonly used objective evaluation indicators: overall accuracy (OA) and Kappa coefficient (Kappa). The OA index describes the proportion of predicted results consistent with the real labels, while the Kappa index starts from the perspective of detection consistency and considers the influence of the number of samples of different categories on detection performance. The OA and Kappa indices are calculated as follows:

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \tag{8}$$

$$\mathrm{Kappa} = \frac{\mathrm{OA}-\mathrm{PRE}}{1-\mathrm{PRE}} \tag{9}$$

$$\mathrm{PRE} = \frac{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})+(\mathrm{TN}+\mathrm{FP})(\mathrm{TN}+\mathrm{FN})}{(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})^2} \tag{10}$$

where TP denotes the number of the correctly predicted changed pixels, TN denotes the number of the correctly predicted unchanged pixels, FP denotes the number of the wrongly predicted unchanged pixels, and FN denotes the number of the wrongly predicted changed pixels.

Besides, the experiments are implemented using the PyTorch platform and the NVIDIA RTX 3060 GPU. The Adam optimizer is used for training in the training phase, and the learning rate is set to 0.0003. The number of training epochs is 100, the batch size and patch size are set to 128 and 7×7, respectively, and the numbers of the attention heads in all attention modules are set to 10. In order to reduce the contingency, all experimental results are taken from the average of five experiments.

B. Comparison Experiments

Since it is hard to acquire the true labels of HSIs, the CD performance of different methods under the condition of a few labeled training samples is particularly important. Therefore, this article first sets a comparison experiment with a low sampling rate, that is, 1% of the samples are randomly selected on the three datasets for training and the same number of samples is used for validation. The remaining samples are used for testing. Certainly, other methods are followed in the same way. Meanwhile, in the prediction results of all methods, white represents changed pixels and black represents unchanged pixels.

To validate the effectiveness of the proposed TriTF, some open-source deep learning-based methods and recent outstanding methods are selected to conduct the comparison experiments, including hybrid spectral CNN

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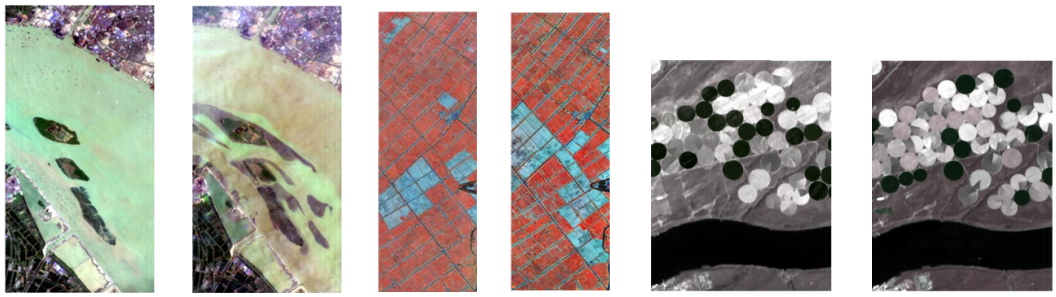


Fig. 7. Pseudo-color images of the three datasets. (a) River in May 3, 2013. (b) River in December 31, 2013. (c) Farmland in May 3, 2006. (d) Farmland in April 23, 2007. (e) USA in May 1, 2004. (f) USA in May 8, 2007.

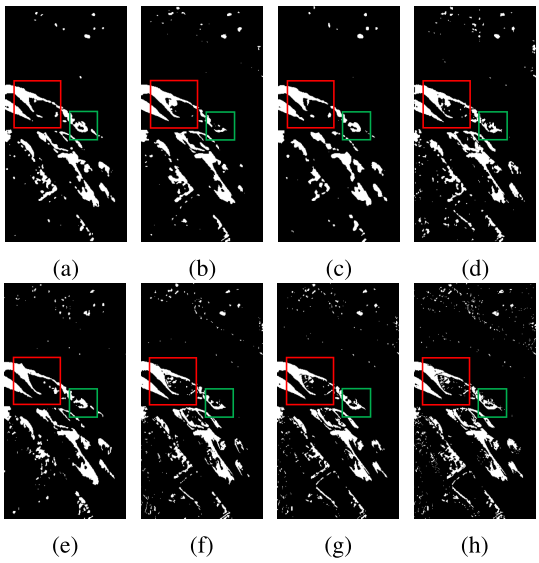


Fig. 8. CD results of different methods on the River dataset. (a) HybridSN. (b) HSI-BERT. (c) DBDA. (d) SSFTT. (e) CSANet. (f) SpectralFormer. (g) TriTF. (h) Ground-truth map.

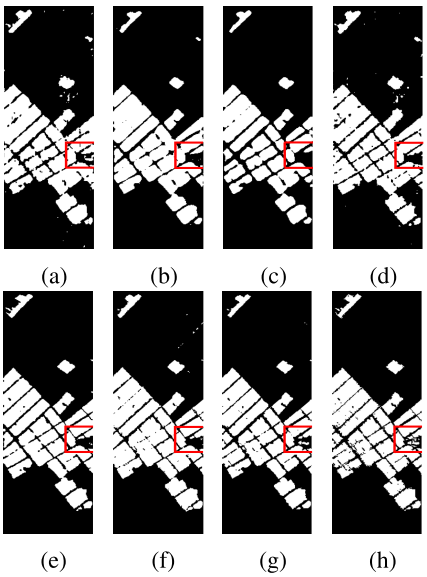


Fig. 9. CD results of different methods on the Farmland dataset. (a) HybridSN. (b) HSI-BERT. (c) DBDA. (d) SSFTT. (e) CSANet. (f) SpectralFormer. (g) TriTF. (h) Ground-truth map.

(HybridSN) [45], double-branch dual-attention (DBDA) mechanism network [46], bidirectional encoder representation from transformers (HSI-BERT) [47], SSFTT [33], CSANet [37], and SpectralFormer [31]. Besides, the comparison methods adopt the same patch size and batch size as TriTF for the experiments. The objective evaluation indicators obtained by different methods are shown in Table II, the highest scores are marked in bold, and the second highest scores are marked with underscores. The red font is the improvement rate of the proposed TriTF compared with the second-best algorithm. Also, the analysis of the experimental results is divided into four parts, namely, the analysis of experiments on each dataset and the summary of these experiments.

1) River Dataset: The prediction results and the ground-truth map are shown in Fig. 8. It can be seen that the prediction results of HybridSN, HSI-BERT, and DBDA have serious clumping phenomenon, so these methods lack the ability to detect fine changes. According to the red square area, there are too many false alarms in the detection result of SSFTT and too many missed alarms in the detection result of CSANet. Also, according to the green square area, the prediction result of TriTF has the highest consistency with the ground-truth map. Obviously, the CD results are consistent with Table II, the proposed TriTF achieves the best detection performance, and the Kappa index obtained by TriTF is 4.21% higher than the second place.

2) Farmland Dataset: The CD results of different methods and the ground-truth map are shown in Fig. 9. It can be seen that the boundary of the changed area is too neat in the CD results of HSI-BERT and DBDA, while the boundary of the changed area in the ground-truth map is rough. Thus, these methods are not effective at identifying class boundaries. Meanwhile, there are many discrete false alarms in the prediction results of HybridSN, SSFTT, and SpectralFormer. According to the red square region, only HybridSN and the TriTF can detect the changed area below. Thus, both the prediction results and Table II show that the TriTF performs better than other methods on the Farmland dataset.

3) USA Dataset: The CD results of different methods and the ground-truth map are shown in Fig. 10. According to the red square areas, it can be seen that the HybridSN, HSI-BERT,

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TABLE II
OA AND KAPPA INDICES OF DIFFERENT METHODS WITH 1% TRAINING SAMPLES

Dataset	Index	Methods / Year							
		HybridSN	HSI-BERT	DBDA	SSFTT	CSANet	SpectralFormer	TriTF	
		2019	2019	2020	2022	2022	2022	-	
River	OA	0.9590	0.9627	0.9530	0.9586	0.9520	0.9678	**0.9743±0.0007**	**↑0.56%**
	Kappa	0.7345	0.7517	0.6944	0.7375	0.6811	0.7941	**0.8362±0.0051**	**↑4.21%**
Farmland	OA	0.9509	0.9561	0.9577	0.9546	0.9650	0.9681	**0.9741±0.0009**	**↑0.60%**
	Kappa	0.8805	0.8948	0.8955	0.8903	0.9150	0.9230	**0.9369±0.0026**	**↑1.39%**
USA	OA	0.9250	0.9508	0.9412	0.9340	0.9386	0.9498	**0.9572±0.0019**	**↑0.64%**
	Kappa	0.7811	0.8563	0.8280	0.8279	0.8222	0.8559	**0.8734±0.0057**	**↑1.71%**

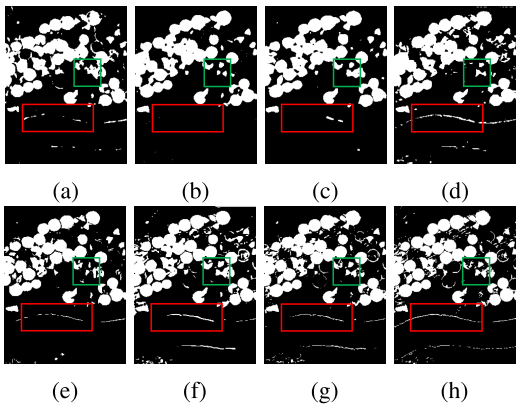


Fig. 10. CD results of different methods on the USA dataset. (a) HybridSN. (b) HSI-BERT. (c) DBDA. (d) SSFTT. (e) CSANet. (f) SpectralFormer. (g) TriTF. (h) Ground-truth map.

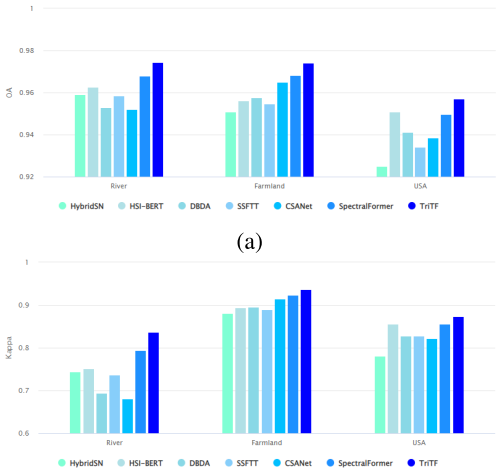


Fig. 11. (a) OA and (b) Kappa indices of different methods on the three datasets with the sampling rates of 1%.

and DBDA methods can hardly identify the changes of river lines, while SSFTT can capture the changes of river lines completely, but there are a lot of false alarms. In all methods, TriTF could capture more changes with a lower false detection rate, and the width of the river line changes is the most similar to the ground-truth map. According to the green square areas, it can be seen that there are masses of missed alarms in SSFTT, HSI-BERT, and CSANet, while the detection result of TriTF is very consistent with the ground-truth map. Besides, as can be seen from Table II, the detection result of TriTF on this dataset is superior to other methods.

4) Summary: To intuitively compare the detection performance of different methods, we draw two bar charts of OA and Kappa indices according to Table II, as shown in Fig. 11. It can be seen that on the three datasets, the TriTF has obvious advantages in both Kappa and OA indices and the advantage is greatest on the River dataset with high detection difficulty (more discrete change pixels and serious class imbalance issue). Therefore, the TriTF has excellent CD performance even though the sampling rate is low, and its dependence on labeled samples is weak.

C. Ablation Study of TriTF

In the proposed TriTF, a transformer structure based on LSA is used as the backbone of the network, and the middle- and long-range information from the BSA branch and the temporal information from the PTA branch are integrated into the backbone together. To prove the effectiveness of the BSA and PTA, we conduct the ablation experiments by taking the network backbone as the baseline and taking different combinations of branches as the contrasts. The contrasts are set as follows: contrasts 1 and 2, respectively, reflect the CD performance of PTA and BSA working alone; contrast 3 is the combination of the baseline and PTA; and contrast 4 is the combination of the baseline and BSA. Meanwhile, other parameters and structures remain unchanged to analyze the validity and necessity of different branches in the network. The results of the ablation experiments are shown in Table III.

TABLE III
QUANTITATIVE COMPARISON OF DIFFERENT BRANCHES

Methods	River		Farmland		USA	
	OA	Kappa	OA	Kappa	OA	Kappa
Baseline	0.9724	0.8196	0.9707	0.9281	0.9253	0.7833
Contrast 1	0.9737	0.8311	0.9724	0.9332	0.9570	0.8731
Contrast 2	0.9728	0.8244	0.9737	0.9366	0.9561	0.8698
Contrast 3	0.9732	0.8252	0.9714	0.9287	0.9559	0.8703
Contrast 4	0.9727	0.8217	0.9714	0.9301	0.9563	0.8705
TriTF	**0.9743**	**0.8362**	**0.9741**	**0.9369**	**0.9572**	**0.8734**

According to Table III, on the River dataset, both BSA and PTA branches could effectively improve the detection performance of the network, and the PTA branch has an obvious

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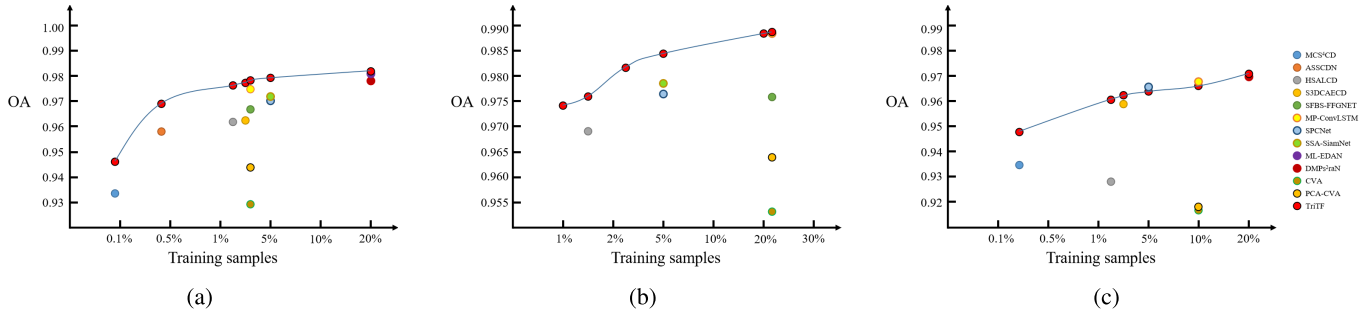


Fig. 12. OA indices of different methods on the three datasets with different sampling rates. (a) River dataset. (b) Farmland dataset. (c) USA dataset.

effect, so TriTF achieved a Kappa improvement of 1.66% compared to the baseline. On the Farmland dataset, contrast 2 performs much better than the baseline, which confirms the importance of the middle- and long-range dependencies, and the CD performance of TriTF relies heavily on the BSA branch. On the USA dataset, the detection performance of the network backbone running alone is much lower than that of joining other branches. Both BSA and PTA can vastly improve the detection performance of the network, and TriTF can achieve a large Kappa increase of 9% compared with the baseline. In conclusion, all these ablation experiments showed that the BSA and PTA in the proposed network can enhance the CD performance of the network, and the best results can be obtained when all branches are used together.

V. Discussion

A. Comparison Experiments With Various Sampling Rates

To make the experiments more complete, more comparison experiments with various sampling rates are conducted to prove the superiority of TriTF, and many superior or SOTA HSI-CD methods in recent two years are selected as the comparing methods, including the HSI-CD based on active CNNs and spatial–spectral affinity graph learning (HSALCD) [48], simplified 3-D convolutional autoencoder (S3DCAECD) [49], MP-ConvLSTM [38], subpixel convolution-based CD network (SPCNet) [50], and DMPs^{2}raN [28]. What needs illustration is that these methods are not open-sourced, so the data of this part of the experiments and the total sampling rate are provided by the original papers of these methods. The OA and Kappa indices obtained by these methods are shown in Table IV, which are ranked according to the sampling rates on the River dataset from low to high.

According to Table IV, from the perspective of each dataset, the proposed TriTF achieves the highest OA and Kappa indices on the River and Farmland datasets no matter whether the sampling rate is high or low. Among them, the advantage of TriTF is more obvious on the River dataset, indicating that TriTF is better at detecting discrete or fine surface changes. On the USA dataset, the CD performance of TriTF is very little affected by the number of training samples. For example, the Kappa index is only improved by about 3%, while the sampling rate is increased from less than 2% to 20%. For this reason, the TriTF has no advantages at the sampling rates of 5% and 9.77%. However, from the perspective of each sampling rate, TriTF is the optimal method under each sampling rate condition.

TABLE IV
OA and Kappa Indices of Different Methods With Different Sampling Rates

Methods	Year	Total training samples	Index	Datasets		
				River	Farmland	USA
HSALCD	2022	2300 / 950 / 1450	OA	0.9620	0.9670	0.9280
			Kappa	0.7430	0.9120	0.7660
TriTF	-	2300 / 950 / 1450	OA	**0.9765**	**0.9759**	**0.9605**
			Kappa	**0.8480**	**0.9415**	**0.8854**
S3DCAECD	2021	3%	OA	0.9626	-	0.9588
			Kappa	0.8031	-	0.8834
TriTF	-	3%	OA	**0.9775**	**0.9816**	**0.9623**
			Kappa	**0.8564**	**0.9552**	**0.8914**
MP-ConvLSTM	2022	3.36% / 20.95% / 9.77%	OA	0.9750	0.9889	**0.9677**
			Kappa	0.8443	0.9723	0.8998
TriTF	-	3.36% / 20.95% / 9.77%	OA	**0.9785**	**0.9892**	0.9660
			Kappa	**0.8599**	**0.9730**	**0.9010**
SPCNet	2022	5%	OA	0.9701	0.9766	**0.9656**
			Kappa	0.8029	0.9430	**0.9055**
TriTF	-	5%	OA	**0.9792**	**0.9846**	0.9638
			Kappa	**0.8641**	**0.9626**	0.8939
DMPs^{2}raN	2022	20%	OA	0.9780	-	0.9696
			Kappa	0.8586	-	0.9124
TriTF	-	20%	OA	**0.9818**	**0.9889**	**0.9708**
			Kappa	**0.8829**	**0.9729**	**0.9161**

For a more intuitive comparison, we draw Fig. 12 according to Table IV and additionally add some outstanding HSI-CD methods in the recent two years, including the semisupervised CD with multilayer cascade screening strategy (MCS^{4}CD) [51], attention-based spatial–spectral fusion network (ASSCDN) [52], SFBS-FFGNET [26], SSA-SiamNet [29], multilevel encoder–decoder attention network (ML-EDAN) [53], the most representative spectral analysis-based method CVA [6], and PCA-CVA [54]. Besides, since the different original papers used different datasets for the experiments, the numbers of comparison methods in Fig. 12(a)–(c) are different. Also, the legend of Fig. 12 is on the right.

B. Impact of Patch Size

In patchwise input networks, the hyperparameter patch size has a great influence on the CD performance and training speed of the network. To this end, different patch sizes are used for experiments at the sampling rate of 1% to analyze the influence of patch size on TriTF. As shown in Fig. 13, when the patch size is set to 3×3, the network detection performance is the worst. When the patch size is set to 5×5, 7×7, or 9×9, the detection performance of the network is similar. Meanwhile, on the River and USA datasets,

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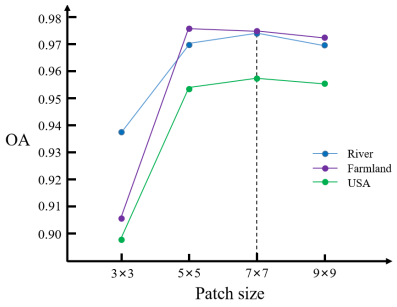


Fig. 13. OA indices of TriTF with different patch sizes.

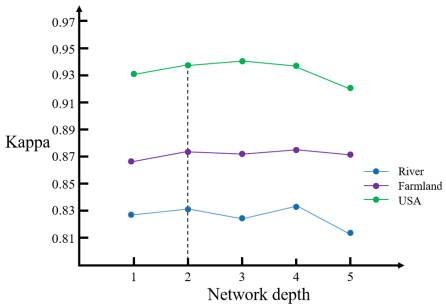


Fig. 14. Kappa indices of TriTF with different network depths.

the network performs best when the patch size is set to 7×7, while the best patch size is 5×5 on the Farmland dataset. In summary, it is most appropriate to set the patch size to 7×7.

C. Impact of Network Depth

As mentioned in Section III-A4, the network depth r is a hyperparameter and it affects the detection performance and complexity of TriTF. Therefore, to analyze the influence of the network depth, the network depth is set as 1–5 to conduct experiments. As shown in Fig. 14, when the network depth increases from 1 to 2, the Kappa index on the three datasets all increases significantly. However, when the network depth continues to increase, the Kappa index is not improved significantly but decreased. On the whole, the detection performance is better when the network depth is set to 2 or 4. Thus, considering the increase in computing amount caused by increasing the network depth, it is best to set the network depth to 2.

D. Comparison of Attention Modules

Currently, many attention mechanisms have been proposed to capture valuable information from different perspectives. Among them, some outstanding modules have been widely used in HSI processing. For example, in HSI-CD tasks, Siamese cross-temporal attention (SCTA) is used to model temporal correlations between bitemporal HSIs [35], [37]; convolutional block attention module (CBAM) is a plug-and-play module that can perform attention operations in both spatial and channel dimensions [55]; and dual attention module (DAM) is composed of position attention module and channel attention module, which can effectively enhance the discriminant ability of feature representation [56]. To demonstrate the validity of our proposed PTA and BSA, the three attention modules above are used to replace our proposed attention modules. In addition, the comparative experiments are divided into two groups according to their working scopes, and the conditions of the experiments are the same as those in Section IV. The experimental results are shown in Table V.

TABLE V
QUANTITATIVE COMPARISON OF DIFFERENT ATTENTION MODULES

Methods	River		Farmland		USA	
	OA	Kappa	OA	Kappa	OA	Kappa
SCTA	0.9709	0.8128	0.9677	0.9218	0.9545	0.8638
CBAM	0.9736	0.8330	0.9705	0.9298	0.9536	0.8662
DAM	0.9722	0.8275	0.9694	0.9275	0.9561	0.8705
TriTF	**0.9743**	**0.8362**	**0.9741**	**0.9369**	**0.9572**	**0.8734**

TABLE VI
COMPLEXITY COMPARISON OF DIFFERENT METHODS ON THE RIVER DATASET

Method	Params(k)	Flops(m)	Time(s)	OA	Kappa
HybridSN	402	**97**	98	0.9590	0.7345
HSI-BERT	**130**	1277	303	0.9627	0.7517
DBDA	377	8467	786	0.9530	0.6944
SSFTT	147	182	**86**	0.9586	0.7375
CSANet	2453	11990	297	0.9520	0.6811
SpectralFormer	329	4339	1563	0.9678	0.7941
TriTF	191	2410	347	**0.9743**	**0.8362**

According to Table V, the CD performance of SCTA is far inferior to TriTF, indicating that the performance of PTA on all three datasets is superior to SCTA. Besides, TriTF has more significant advantages on the River dataset, which is more difficult to detect due to the class-imbalanced issue. Thus, PTA can make better use of temporal dependencies to improve the network to extract the change information. As for CBAM, its performance is close to TriTF on the River and Farmland datasets but poor on the USA dataset. The CD performance of DAM is between SCTA and CBAM, and it performs well on the USA dataset. Overall, the proposed TriTF achieved the highest OA and Kappa indices on all three datasets, indicating that the proposed PTA and BSA outperform other attention modules and can better supplement the deficiency of local feature extraction in HSI-CD tasks.

E. Complexity Analysis

In addition to the detection accuracy, complexity is also a key factor affecting the practicability of the CD method. To this end, this article analyzes the complexity of the experiments in Section IV-B. In the experiment, the hyperparameters patch size and batch size, which affect the complexity of the method, are uniformly set as 7×7 and 128 to control variables, respectively. Table VI shows the comparisons of the different methods.

According to Table VI, the HSI-BERT, SSFTT, and our proposed TriTF have a small number of params, while CSANet and HybridSN have too many params. In terms of the floating point of operations (flops), HybridSN, SSFTT, and HSI-BERT perform better. Consistent with the previous two indicators, the total running time of HybridSN and SSFTT is also much shorter than other methods, indicating that the computational complexity of these two methods is quite low. On the contrary,

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DBDA and SpectralFormer have a much higher complexity than other methods. As a whole, our proposed TriTF is at a moderate level in terms of complexity but has significant advantages over other methods in terms of detection accuracy. Therefore, our proposed TriTF has better effectiveness and availability.

VI. Conclusion

In this article, a novel TriTF is proposed for HSI-CD. The proposed framework can utilize many different types of information in HSIs to improve the CD performance. In TriTF, PTA utilizes the temporal dependencies in HSIs to propagate valuable information from original HSIs to the difference-temporal image; BSA captures the middle- and long-range dependencies of land cover and achieves the long-range feature propagation between land cover, which is unconstrained by the receptive field. Meanwhile, the LSA in the transformer-based backbone models the local correlations and extract features in the range of patches. The effectiveness of TriTF is verified from different aspects, and abundant experiments prove that TriTF can achieve the SOTA CD performance when the training samples are sufficient or insufficient and has competitive computational efficiency.

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Xianghai Wang was born in 1965. He received the Ph.D. degree from Jilin University, Changchun, China, in 1999.

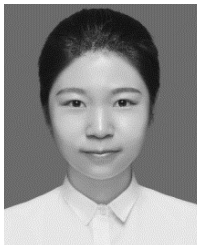
He is currently a Professor with the School of Geography and the School of Computer and Information Technology, Liaoning Normal University, Dalian, China. His research interests include computer graphics, multimedia information processing, and hyperspectral image.

Dr. Wang is a Senior Member of the China Computer Federation (CCF).



Keyun Zhao was born in 1997. He is currently pursuing the M.S. degree with the School of Computer and Information Technology, Liaoning Normal University, Dalian, China.

His research interests include remote sensing image processing and mathematical modeling.



Xiaoyang Zhao was born in 1994. She received the M.S. degree from the College of Computer and Information Technology, Liaoning Normal University, Dalian, China, in 2019. She is currently pursuing the Ph.D. degree with the School of Geography, Liaoning Normal University.

Her research interests include remote sensing image processing and deep learning.



Siyao Li was born in 1999. She is currently pursuing the M.S. degree with the School of Computer and Information Technology, Liaoning Normal University, Dalian, China.

Her research interests include remote sensing image processing and deep learning.

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